An Investigation of TID Mitigation for Analog Inference Accelerators

Neal Bagai, Brian Sierawski

VANDERBILT UNIVERSITY

Abstract

Analog In Memory Computing (AIMC) arrays are an energy efficient method to perform deep neural network (DNN) inferencing tasks. However, their performance in a space environment can be compromised by exposure to total ionizing dose (TID). This study investigates array configurations made to increase the resilience of inferencing tasks on AIMC devices exposed to TID.

Introduction and Background

- Interest has increased in using AI models to perform inferencing locally on vehicles in space, avoiding a lengthy data transmission to the ground for processing.
- Limited power budgets of spacecraft prevent traditional digital AI accelerators from being utilized.
- AIMC "crossbar" arrays perform matrix-vector multiplication (MVM), the core math operation in DNNs, by representing operands as physical analog quantities.
- DNNs tolerance to the inherent errors in the analog results of AIMC, makes AIMC uniquely adept at lower power AI tasks.
- The space environment introduces total ionizing dose, resulting in a gradual shift of memory cell programmed currents, and affecting accuracy during inferencing tasks.

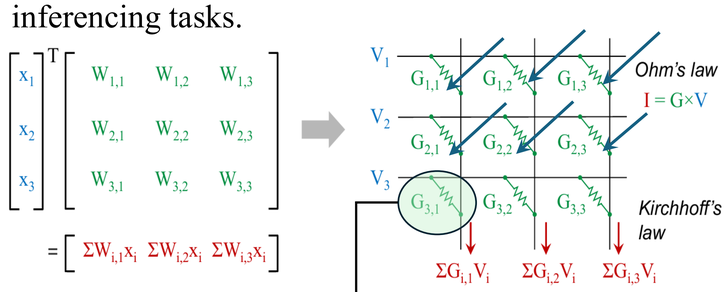

Figure 1. Matrix vector multiplication values and corresponding to AIMC array physical quantities [2].

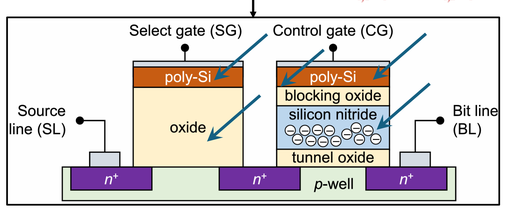

Figure 2. 2T SONOS memory cell used in the crossbar array [1]. *Radiation sources denoted with blue arrows.*

Methodology

- An AIMC circuit utilizing SONOS charge-trap memory cells was simulated to run the ResNet-20 residual network on the CIFAR-10 image dataset using CrossSim, created by Sandia National Laboratories.
- A custom model for TID response of the SONOS memory cells was created from experimentally derived data on irradiated SONOS crossbar array [1].

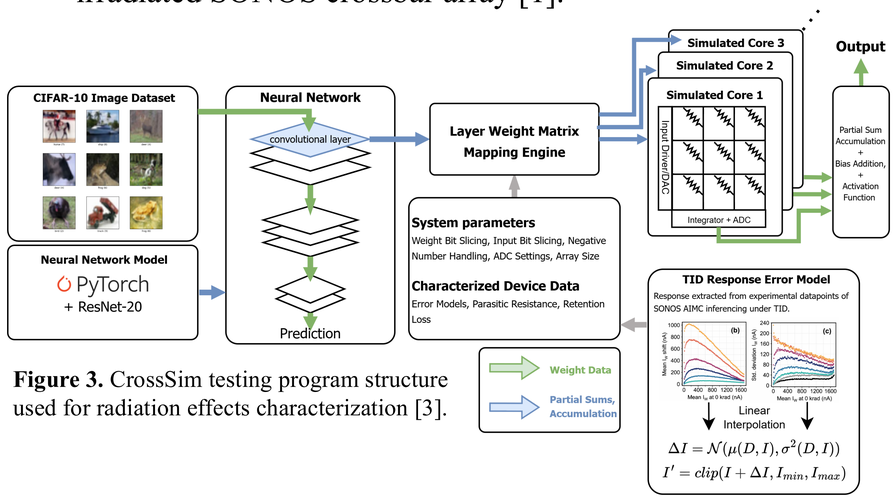

Figure 3. CrossSim testing program structure used for radiation effects characterization [3].

Different crossbar array configurations were tested by sweeping the following device parameters through a range of values [3].

- Nslices: the number of memory cells comprising a layer weight on the physical crossbar array.
- Input_bit_slices: The number of bit slices comprising the input value of the MVM. A value > 1 uses a multiplexer to separate the input bits.
- alpha_mu and alpha_sig: gain values applied to the two components of TID response error model.
- TID Amount: Irradiating dose on range of 0-1500 krad(Si).

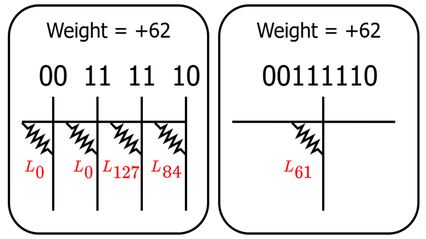

Figure 4. A crossbar array configured with weight bit slicing. (2 bits/cell & 8 bits/cells)

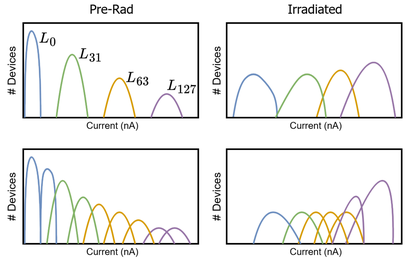

Figure 5. Series of histograms of programmed current (I) with and without bit slicing. *Shapes simplified.*

Results

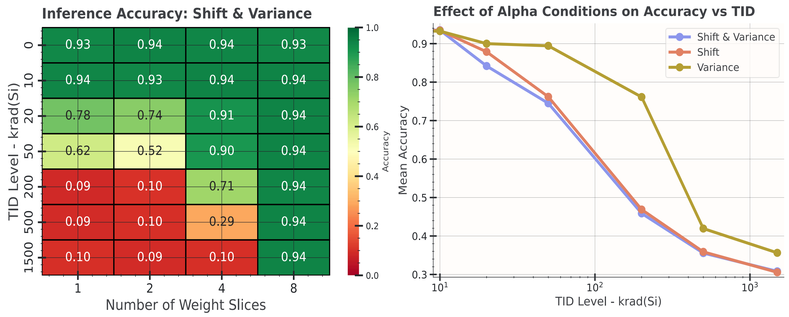

Figure 7. & 8. Accuracy results from inference tasks across TID range. Left figure is a heatmap based on number of weight bit slices. Right figure based on changes to components of TID error model.

- In the TID response error model, the current shift (μ_{I_w}) component had greater effect than current spread (σ_{I_w}) on the degradation of DNN classification performance.
- DNN classification accuracy remained high (>80%) for configurations with higher weight bit slicing (>4 cells per weight) up to TID levels of ~50-100 krad(Si).

Conclusion

- While slicing stored weight values is an effective method to increase the robustness of DNN inferencing results under TID, there is significant a tradeoff with increased physical array size and power consumption.
- Variance of (I_w) contributes less to inference accuracy degradation than mean shift of (I_w) away from initial levels. Suggests alternate mitigation strategies, such as weight refreshing and dynamic error correction [1].
- Criteria for power and size constraints should be considered before adding weight bit slicing to mitigate accuracy degradation under TID.
- Ongoing investigation for hybrid weight mapping setups.

References

[1] T. P. Xiao *et al.*, "Total Ionizing Dose Response of 128 Analog States in Computational Charge-Trap Memory," *IEEE Transactions on Nuclear Science*, vol. 71, no. 4, pp. 446–453, Apr. 2024, doi: 10.1109/TNS.2023.3333812.

[2] T. P. Xiao *et al.*, "On the Accuracy of Analog Neural Network Inference Accelerators," *IEEE Circuits and Systems Magazine*, vol. 22, no. 4, pp. 26–48, 2022, doi: 10.1109/MCAS.2022.3214409.

[3] T. Xiao, C. Bennett, B. Feinberg, M. Marinella, and S. Agarwal, "CrossSim Inference Manual v2.0," SAND2022-7074R, 1869509, 706724, May 2022. doi: 10.2172/1869509.

Acknowledgments

This work was supported in part by the 2025 Vanderbilt Summer Research Program.